

where we used the coarea formula in the first inequality [Fed59], and the fact that the map from $\hat{\mathcal{D}}_r(\Omega) \cap S_t$ to $(\hat{V}_{\hat{j}}, \angle_{Tits})$ defined by $\sigma(t) \mapsto [\sigma]$ is locally $\pi/(2C_k t)$ -Lip due to (23); see for instance [BBI01, Proposition 1.7.8].

To conclude, we note that $\mathcal{H}^{n-1}(\hat{V}_{\hat{j}})$ has a positive lower bound independent of $\hat{j}$, and thus of Ω . Indeed, because of the volume growth assumption on each end of M , all the connected components of M_∞ are Alexandrov $CBB(0)$ spaces of dimension $n-1$; see Lemma 3.1, (4). Accordingly, all the open sets $\hat{V}_j$, $j = 1, \dots, J$, have strictly positive $(n-1)$ -dimensional Hausdorff measure, so that $\min_j \mathcal{H}^{n-1}(\hat{V}_j) > 0$. □

Remark 4.6. In the stronger assumptions of Corollary 1.4, that is when the sectional curvature of M is non-negative outside some compact set K of M , there is no need to conformally deform the metric of M . In particular, one can replace $\varphi \equiv 1$ and $f \equiv 0$ in the above proof.

An even simpler case is when M has nonnegative sectional curvatures outside a compact set $K \subset M$, M has only one end, and $\text{diam } M_\infty < \pi$. Indeed in this case, we *claim* that there exists R_0 large enough so that any minimizing geodesic connecting two points of $M \setminus B_{R_0}(o)$ does not intersect K . One can thus implement Brendle's approach to prove the isoperimetric inequality on $M \setminus B_{R_0}(o)$. Namely, it is enough to consider $\hat{A}_r := \{y \in A_r : \Phi_r(y) \notin B_{R_0}(o)\}$, so that the minimizing geodesics from $y \in \hat{A}_r$ to $\Phi_r(y)$ is totally contained in $M \setminus K$, and the curvature is non-negative along that geodesic. Since removing $B_R(o)$ from the image of Φ_r does not affect the asymptotic volume growth, one can conclude the proof.

To show the above claim, suppose there exist two sequences of points $\{x_j\}$, $\{y_j\}$ such that $\text{dist}(o, x_j) > j$, $\text{dist}(o, y_j) > j$ and a unit-speed minimizing geodesic γ_j from x_j to y_j which intersects K . On the one hand, $\text{dist}(x_j, y_j) \geq 2j - 2\text{diam } K$. On the other hand $\lim_{j \rightarrow \infty} \text{dist}(x_j, y_j)/j < 2$ by Lemma 3.1 (2) and the assumption on M_∞ , thus giving a contradiction.

A similar argument permits also to deal the case in which there are several ends, but each connected component of M_∞ has diameter $< \pi$.

Remark 4.7. According to the previous remark, when all the connected components of M_∞ have diameter $< \pi$ the isoperimetric constant in (9) restricted to $C_c^\infty(E \setminus B_{R_0}(o))$, where E is an end of M , is the sharp Euclidean one, i.e. $n\omega_n^{1/n}\theta^{1/n}$ with $\theta = \limsup_{r \rightarrow \infty} \omega_n^{-1}r^{-n} \text{vol}(B_r(o) \cap E)$ the AVR of E . Clearly, the constant in (9) for general functions in $C_c^\infty(M)$ necessarily depends on the whole manifold and can not be estimated only in term of the quantities appearing in our assumptions. For instance, adding a tubular handle $[0, L] \times \epsilon \mathbb{S}^{n-1}$ to M near o makes the isoperimetric constant explode when $L \rightarrow \infty$ without adding negative curvature outside a compact neighborhood of o .

5. SUBMANIFOLDS IN AMBIENT MANIFOLDS WITH NONNEGATIVE SECTIONAL CURVATURE OUTSIDE A COMPACT SET

Throughout this section we will assume that (M^{n+m}, g) is a complete smooth $(n+m)$ -dimensional Riemannian manifold satisfying

$$\text{Sect}^M \geq 0$$

outside some compact set $K \subset M$. Moreover, we will assume that, given a reference point $o \in M$, AVR > 0 on each end E_j of M , i.e.

$$\theta_j = \limsup_{r \rightarrow \infty} \frac{\text{vol}(B_r^M(o) \cap E_j)}{r^{n+m}} > 0.$$

Let Σ be a compact submanifold of dimension n . We will denote by $\bar{\nabla}$ the Levi-Civita connection on M and by ∇^Σ the induced connection on Σ . The second fundamental form II of Σ is given